

ECON 0100 | Homework B6 | International Trade

I Solutions

Answers and work are shown in the red blocks.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | {++Opening the Market++}

The Ministry of Magic hired an external committee of advisors ~~to find means by which to raise government revenue without raising taxes.~~ One classically trained magical economist proposed opening the closed domestic butterbeer market to the less expensive global market ($P = 40$) ~~while imposing an import tariff of 2 galleons.~~ Ministry officials were pleased with the option but also raised concerns that domestic butterbeer sellers would be harmed by the lower price. {++Use a graph and algebra to analyze the impact.++}

- a) Imports or exports? **imports**, exports
- b) Quantity demanded: **120** Quantity supplied: **40** Imports: **80**
- c) Consumer surplus before: **1600** after: **3600**

d) Producer surplus before: **1600** after: **400**

e) Change in total surplus: **+800**

f) Buyers: **gain**, lose Sellers: gain, **lose**

Closed market: 80 bottles at 60 galleons, CS = PS = 1600.

At the world price of 40, below the domestic price, buyers buy abroad: the market **imports**.

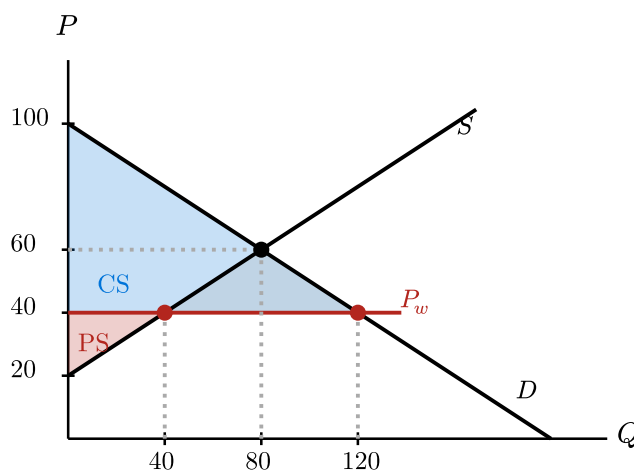
Quantity demanded 120, quantity supplied 40; imports $120 - 40 = 80$.

CS = $\frac{1}{2} \cdot 120 \cdot (100 - 40) = \mathbf{3600}$ (+2000).

PS = $\frac{1}{2} \cdot 40 \cdot (40 - 20) = \mathbf{400}$ (-1200).

Total surplus rises by **800**, the triangle $\frac{1}{2} \cdot 80 \cdot |60 - 40|$ between the curves and the world price. Buyers gain more than sellers lose.

The officials were right: sellers lose 1200. But buyers gain 2000, so the market as a whole gains 800.



Q2 | An Alternative World

If instead of a price of 40 galleons in Question 1 the global price had been 80 what would have been the distributional impact of opening the domestic market for trade? {++Use a graph and algebra. ++} ~~No need to do math here. Simply use a graph to show the impact on the market.~~

a) Imports or exports? imports, **exports**

b) Quantity demanded: **40** Quantity supplied: **120** Exports: **80**

c) Consumer surplus before: **1600** after: **400**

d) Producer surplus before: **1600** after: **3600**

e) Change in total surplus: **+800**

f) Buyers: **lose** Sellers: **gain**, lose

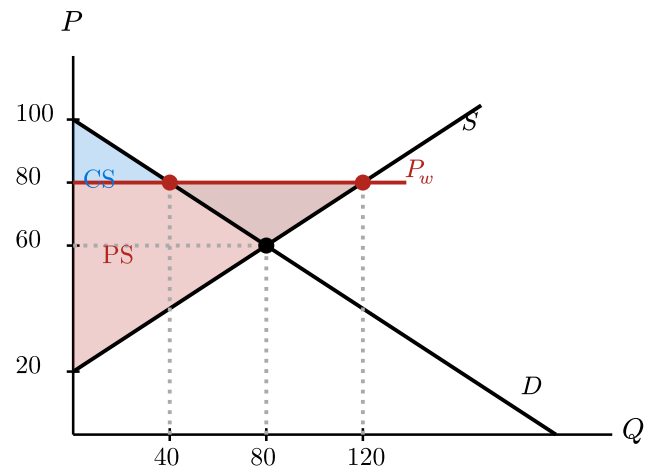
Closed market: 80 bottles at 60 galleons, $CS = PS = 1600$.

At the world price of 80, above the domestic price, sellers sell abroad: the market **exports**.

Quantity demanded 40, quantity supplied 120; exports $120 - 40 = 80$.

$CS = \frac{1}{2} \cdot 40 \cdot (100 - 80) = 400$ (-1200). $PS = \frac{1}{2} \cdot 120 \cdot (80 - 20) = 3600$ ($+2000$).

Total surplus rises by **800**, the triangle $\frac{1}{2} \cdot 80 \cdot |60 - 80|$ between the curves and the world price. Sellers gain more than buyers lose.



Q3 | {++A Changing World Price++}

{++Return to Q1. Suppose the global price rises from 40 to 50 galleons.++}

a) Quantity demanded: **100** Quantity supplied: **60** Imports: **40**

b) Consumer surplus: **2500** Producer surplus: **900**

c) Imports have: risen, stayed the same, **fallen**

At 50, buyers want 100 and domestic sellers supply 60: imports fall from 80 to **40**.

$CS = \frac{1}{2} \cdot 100 \cdot (100 - 50) = 2500$, down from 3600; $PS = \frac{1}{2} \cdot 60 \cdot (50 - 20) = 900$, up from 400. The gain from trade shrinks from 800 to 200: the closer the world price is to the domestic price, the less there is to gain.